

Dataset	Metric	SasRec(BCE)	SasRec(CE)	SasRec(SCE)	CL4SRec	DuoRec	EC4SRec	BT-SR(Ours)	Improve
ML 1M	hr@1	0.0160	<u>0.0213</u>	0.0200	0.0182	0.0200	0.0213	0.0240* $\pm$ 0.0048	+12.7%
	hr@5	0.0493	<u>0.0600</u>	0.0587	0.0476	0.0480	0.0534	0.0639* $\pm$ 0.0024	+6.5%
	hr@10	0.0893	0.0847	<u>0.0933</u>	0.0834	0.0800	0.0880	0.0945* $\pm$ 0.0080	+1.3%
	ndcg@5	0.0319	<u>0.0404</u>	0.0393	0.0342	0.0353	0.0364	0.0436* $\pm$ 0.0020	+7.9%
	ndcg@10	0.0443	0.0494	<u>0.0503</u>	0.0464	0.0456	0.0487	0.0534* $\pm$ 0.0037	+6.2%
	ndcg@50	0.0799	<u>0.0818</u>	0.0800	0.0712	0.0723	0.0748	0.0833* $\pm$ 0.0033	+1.8%
	cov@1	0.0320	<u>0.0887</u>	0.0961	0.0335	0.0361	0.0387	0.0816 $\pm$ 0.0035	-15.1%
	cov@5	0.1473	<u>0.2265</u>	0.2568	0.0969	0.0936	0.1002	0.2223 $\pm$ 0.0083	-13.4%
	cov@10	0.2434	0.3185	0.3645	0.1329	0.1459	0.1591	<u>0.3189</u> $\pm$ 0.0123	-12.5%
Beauty	hr@1	0.0179	0.0269	<u>0.0305</u>	0.0292	0.0305	0.0303	0.0326* $\pm$ 0.0013	+6.9%
	hr@5	0.0538	<u>0.0591</u>	0.0582	0.0555	0.0573	0.0574	0.0606* $\pm$ 0.0025	+2.5%
	hr@10	0.0789	0.0896	<u>0.0905</u>	0.0830	0.0860	0.0815	0.0937* $\pm$ 0.0019	+3.5%
	ndcg@5	0.0355	0.0425	0.0442	0.0412	0.0436	<u>0.0455</u>	0.0463* $\pm$ 0.0002	+1.8%
	ndcg@10	0.0434	0.0524	<u>0.0543</u>	0.0530	0.0527	0.0537	0.0569* $\pm$ 0.0001	+4.8%
	ndcg@50	0.0659	0.0777	0.0794	0.0833	0.0789	0.0794	<u>0.0813</u> $\pm$ 0.0035	-2.4%
	cov@1	<u>0.0620</u>	0.0546	0.0499	0.0603	0.0607	0.0679	0.0491 $\pm$ 0.0078	-20.7%
	cov@5	<u>0.2200</u>	0.1782	0.1760	0.2212	0.2100	0.2103	0.1532 $\pm$ 0.0357	-29.2%
	cov@10	0.3448	0.2752	0.2766	<u>0.3362</u>	0.3234	0.3236	0.3234 $\pm$ 0.0612	-30.5%
Kindle Store	hr@1	0.0451	0.0471	<u>0.0533</u>	0.0469	0.0471	0.0505	0.0564* $\pm$ 0.0007	+5.8%
	hr@5	0.0832	0.0813	<u>0.0899</u>	0.0813	0.0762	0.0785	0.0903* $\pm$ 0.0018	+0.4%
	hr@10	0.0990	0.0965	<u>0.1022</u>	0.0910	0.0897	0.0955	0.1058* $\pm$ 0.0012	+3.5%
	ndcg@5	0.0655	0.0652	<u>0.0726</u>	0.0635	0.0625	0.0652	0.0747* $\pm$ 0.0009	+2.9%
	ndcg@10	0.0705	0.0700	<u>0.0765</u>	0.0662	0.0668	0.0672	0.0796* $\pm$ 0.0005	+4.1%
	ndcg@50	0.0786	0.0803	<u>0.0852</u>	0.0731	0.0771	0.0736	0.0883* $\pm$ 0.0007	+3.6%
	cov@1	0.0396	0.0337	<u>0.0409</u>	0.0363	0.0356	0.0369	0.0440* $\pm$ 0.0005	+7.6%
	cov@5	0.1278	0.1076	<u>0.1492</u>	0.1077	0.1123	0.1082	0.1665* $\pm$ 0.0019	+11.6%
	cov@10	0.1920	0.1665	<u>0.2346</u>	0.1598	0.1719	0.1703	0.2673* $\pm$ 0.0021	+13.9%

Table 1: Metrics with mean and std by 5 runs

Dataset	Users	Items	Interactions	Density
ML-1M	6,040	3,952	1,000,209	4.19%
Kindle Store	23,684	96,830	1,256,065	0.05%
Yelp	42,429	137,039	2,294,804	0.04%
Gowalla	27,516	173,511	2,627,663	0.06%
Beauty	22,363	12,101	198,502	0.07%

Table 2: Statistics of the datasets after preprocessing

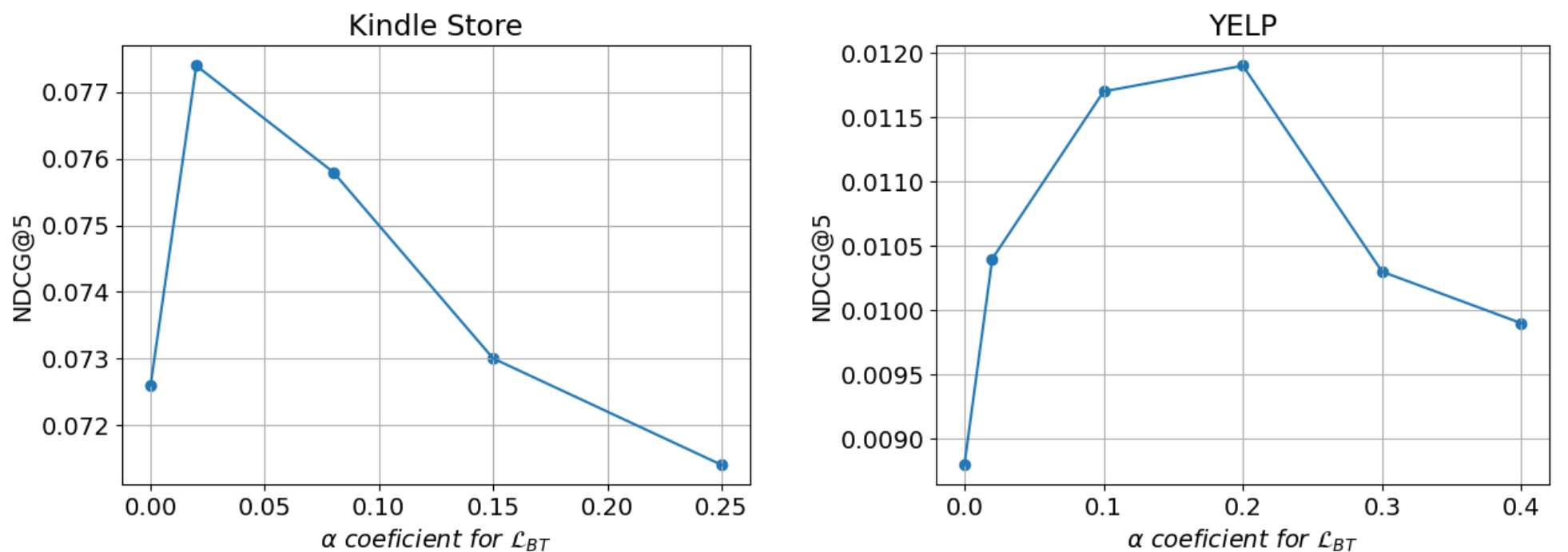


Figure 1: Parameter sensitivity of α for Barlow Twins loss for 2 datasets

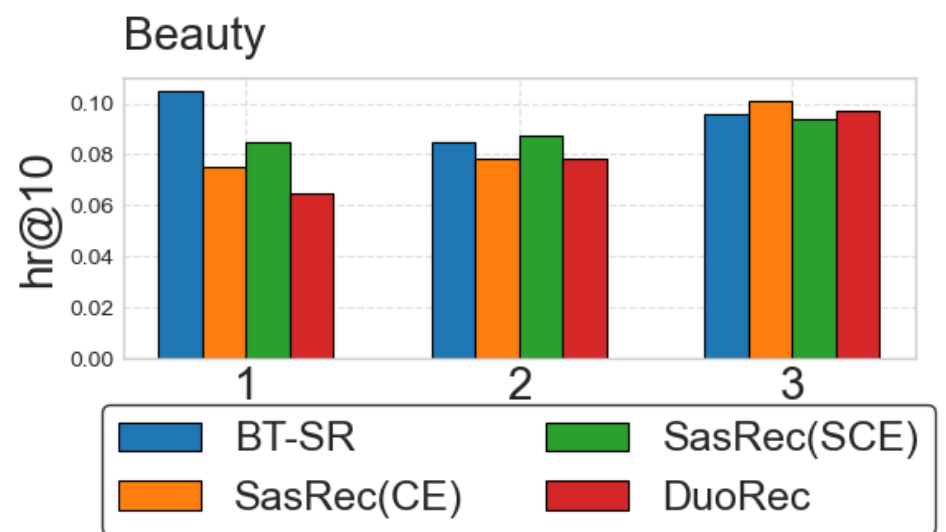
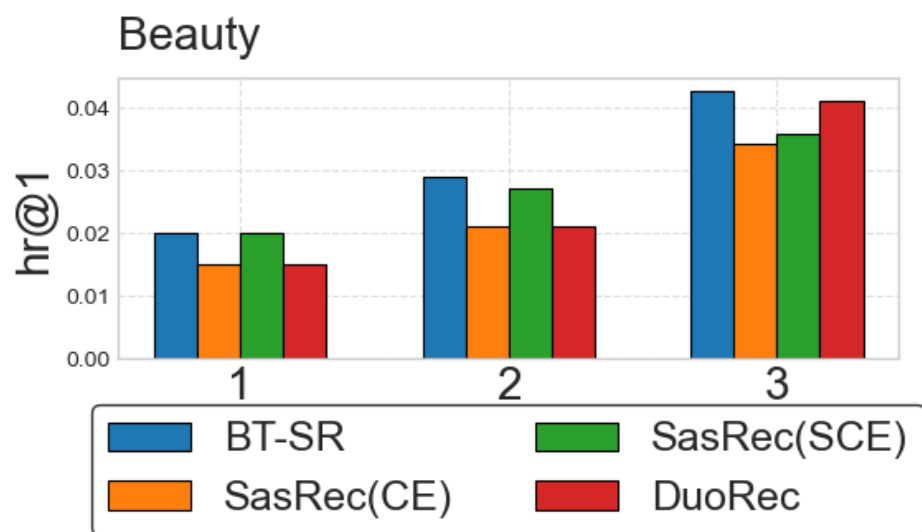


Figure 2: Bucket utility metrics($HR@K$) for Beauty dataset

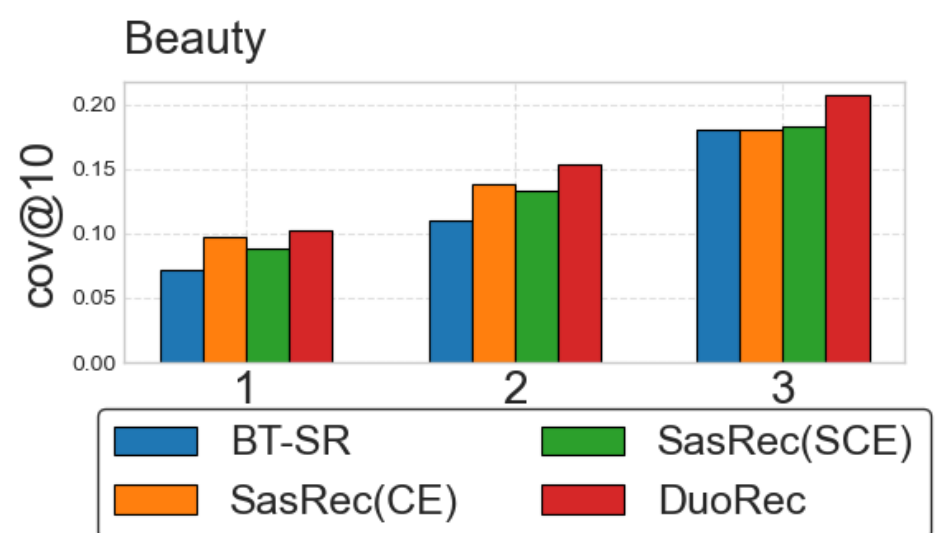
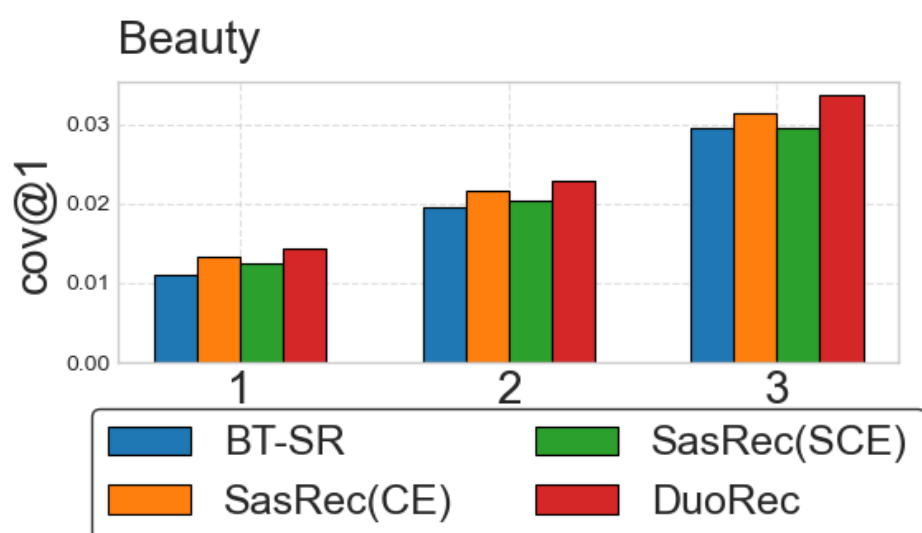


Figure 3: Bucket coverage metrics for Beauty dataset